The New Liquid Gold? Semiconductor Manufacturers, AI Infrastructure, and Modern Markets (Working Title)

DS 190 Capstone — 1st Outline

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Introduction

My introduction frames semiconductors as strategic intermediate goods, similar to oil, that anchor the physical infrastructure of the AI boom. My research question examines how expanding AI data-center demand impacts semiconductor producer prices (PPI) and whether this relationship shifted post-2021. Using global data-center electricity demand to illustrate AI expansion, alongside interest rates and a post-2021 interaction term as controls, the paper outlines a full analytical workflow from time-series regression to sensitivity analysis, strictly distinguishing statistical association from causation.

The Complex Global Chip Production Process & Network

This section will explain why semiconductor production is unusually complex and why the supply of chips cannot be understood as the product of a single firm or country. TLDR: Semiconductor manufacturing involves several interconnected stages, including chip design, fabrication, precision equipment, specialty gases and chemicals, packaging, testing, memory production, and final assembly, with different parts of this process concentrated among specific firms and geographic regions. Companies such as TSMC, Samsung, SK Hynix, and Micron therefore occupy strategically important positions within this network, especially as AI-related demand places additional pressure on already specialized production systems. Ultimately, this section is meant to create a captivating narrative that drives the rest of my project and presentation.

Data and Methodology

This will be the primary empirical and analytical section of my capstone project. My meta-analysis will begin with what I believe are the core variables needed to better quantitatively understand the overarching context previously posed. These variables include monthly semiconductor PPI from the FRED dataset, global data-center electricity demand from the International Energy Agency (IEA) dataset, interest rates from the World Bank and the International Monterrey Fund, and a post-2021 indicator in my time series that is meant to indicate price effects after the "AI Boom" respectively. I chose to use a time-series framework for my project because it accounts for volatile price hikes during the "AI Boom" and observations that might illustrate developments that led up to it.

Given the literature that I have already read on this topic, I hypothesize that there will be a noticeable hike in demand and prices (PPI) from 2018–2019, followed by an even larger increase in these two variables post-2021. From 2018–2019, there was a semiconductor chip shortage as firms and individuals increasingly used these chips for cryptocurrency mining, specifically Ethereum mining. I then expect there to be a slight dip during the COVID-19 pandemic, followed by the larger increase associated with the "AI Boom."

Why Semiconductor Chip Manufacturers Drive the AI Boom

This section will connect the statistical findings back to the larger economic argument that semiconductor manufacturers have become increasingly important actors within the AI economy because they control many of the physical inputs that make AI expansion possible. Additional implications related to this argument will also be included here.

Limitations and Conclusion

My limitations and conclusion section will include any persistent challenges I faced that either delayed my project, caused initial issues in my capstone, or changed the scope or direction of my project entirely.

Bibliography

This section will contain all of the sources I used for this project. There will also be a `.bib` file/document made available, which will also be included in my repository.

Appendix

The appendix will consist of several components, including my code base (R, SQL, Python, etc.), potential datasets I used and/or generated, my Git repository, and other supporting materials.